Footprint borders facilitator guide

Mathematical overview

Object. A border is the ideal infinite repetition in both directions of a finite motif arrangement, bounded across the strip. Only footprints and their orientations count. Ends, guide lines and temporary labels do not. The intended examples are discrete, nonempty patterns with a shortest positive translation P along the border. Finite scraps alone do not establish an infinite symmetry.

Glide theorem. Write $H(x,y) = (x,-y)$ for reflection in the middle line and $T_a(x,y) = (x+a,y)$ for a slide along it. They commute. A glide $G_a = T_a H$ with $a \neq 0$ satisfies $G_a^2 = T_{2a}$. Thus a matching 3 cm glide forces a matching 6 cm slide, but need not force a 3 cm slide or the bare reflection H.

Shortest distances. If a border has a glide along this line but H itself fails, its shortest glide length is P/2. Every glide displacement is $(k + 1/2)P$, k an integer. If H is a symmetry instead, matching glide displacements are kP; the shortest nonzero glide length is P. These statements require a primitive translation P and a fixed glide axis; an arbitrary matching glide need not be the shortest.

Forced motions. Reflections in two perpendicular lines compose to a half-turn about their crossing. The printed crossing lines are perpendicular. For general crossing lines the rotation angle is twice their angle; a half-turn is not a theorem for arbitrary crossing lines. Every composition or inverse of symmetries is a symmetry.

Limits. There are seven discrete frieze symmetry types, but these problems do not prove or require that classification. A simple motif can accidentally add symmetries. The specified notched F-like motif has no nonidentity planar isometry, so its handedness and direction distinguish candidate moves.

What children are meant to discover

- **K-1:** Build repeating borders and use a copy to distinguish slides, flips, glides and half-turns. A local matching attempt is evidence to investigate the repeated rule.
- **Grades 2-3:** Find a glide that works although the flip alone fails; compare shortest distances and change which motions a border permits.
- **Grades 4-5:** Explain forced symmetries by composition and justify a full symmetry description for one chosen construction. Sampling several moves is not a proof of completeness.

Status. Current packets (v4, October 23, 2026): K-1 has seven problems on five pages, Grades 2-3 six on four pages and Grades 4-5 six on five pages. Mathematical examples are verified; the card kit, physical handling and classroom pacing are untested.

Preparation

Materials and readiness

For eleven children, use three adult-led tables. Three-child groups can rotate builder, tester and observer. Print one sheet of **week-35-materials.pdf** per table on card stock at 100%; each half is a set of 24 footprint cards for one pair or trio. Per pair also supply four paper strips at least 70 by 280 mm, two tracing strips, a pencil or washable marker, and a ruler. Provide a 600 mm joined demonstration strip per table. Keep repeated motifs separated and use at least three or four visible repeats.

Motifs: each card carries the notched F-like outline of the introduction, 26 mm wide, with its direction arrow. A set has 12 cards of A and 12 of its mirror over a horizontal line. Turning an A card halfway round gives its half-turn; turning a mirror card gives the vertical mirror. Cards are 29 by 26 mm, so Border E's 3 cm spacing fits; recipe positions on page 4 are card centres. Card edges and blank backgrounds are handling aids, not pattern data. Use tracing paper to reflect a whole border. Adults cut the cards, not children during the investigation.

Boards, tape and tracing: children build each border in a printed box (17.65 by 6 cm), which holds two and a half to three repeats of a 6 cm border with card centres 1.5 cm above and below the middle line. Move to a long strip only when one repeat is longer than 6 cm, such as K-1 Problem 4's longer slide or a Grades 2-3 Problem 6 edit with a 12 cm repeating part. When a border is finished, tack its cards down with removable tape so that testing cannot nudge a card into a match. Lay a tracing strip over the box and always trace the middle line first: the flip is made over it. A rough outline of each footprint with its arrow is enough, since the arrow's place and direction tell the four forms apart. At K-1 the adult traces and the children move the copy.

Preparation estimate: about 20 minutes to cut four sets with scissors (96 cards on straight lines; less with a paper trimmer). The cards are reusable. A set holds any two of page 4's borders on the printed boards, or one on a long strip. Before class, check the 10 cm scale line, test one reflected tracing strip, and rehearse the common launch. The kit is physically unrehearsed.

Readiness: children need to extend a repeating arrangement and compare a moved copy with every motif in the original. Reading and measured lengths can be adult-supported; the youngest can compare slide lengths directly. If keeping the original fixed or mentally continuing the ends is difficult, spend the session on one border and slide matching.

Running the session

Common launch

Let children handle pieces first. At the K-1 table, start with one repeating row in a box and a tracing copy: slide the copy to a matching position and a position that fails. Keep the original fixed. Say aloud the three rules the K-1 page leaves to the adult: the border repeats forever both ways; only footprints count, not strip ends or guide lines; and a slide must move the copy. Introduce the single-motif flip-then-slide visual only before the first glide task. Keep the label A with the same piece during the demonstration. This teaches the move without giving a target border. Compare two or three interior repetitions; explain that the printed ends are a window on an endless pattern. Demonstrate a half-turn about a chosen point when first needed, including a point between motifs.

Session plan

The hour opens with about five minutes of running around. Minutes below count from the launch; the tables have about 35-40 minutes, then a few minutes of sharing. **0-5:** handling and launch. **5-18:** two different shortest slides, then the glide challenge. Let a successful border stay taped in one box while a second is built. **18-32:** a selected reflection or half-turn construction or edit. **32-40:** partner testing and one explanation of a forced motion. **Sharing:** each table shows one border and its generating rule; note what children could actually justify. If attention flags, act out a slide and a half-turn with a hand-held copy as a short movement break.

A good stopping point is one border with a well-defended unusual symmetry. Upper Problem 6 can occupy an entire later discussion. Offer hints only when needed: “Where must this one footprint land?” then “Does its direction agree?” then “What happens if you do your matching move twice?”

Checking a child's own border

The borders on page 4 are witnesses, not the only solutions. Ask for the repeating part and test its seams as well as its interior. A drawing with three locally similar motifs may fail at a join. A complete argument uses a generating rule and shows where every kind of motif lands. When excluding a motion, a single unavoidable mismatch is enough. Do not infer the shortest period from the length of the block the child happened to draw.

Concrete answer bank

A precise way to read these examples

Let the middle line be y = 0. Let **A** be the printed asymmetric footprint, **H A** its reflection across a horizontal line, **V A** its reflection across a vertical line, and **R A** its half-turn. These names describe the entire footprint, including its direction arrow. Positions below are footprint centres (card centres) in centimetres. Take h = 1.5 cm. Footprints should be 25-30 mm wide: smaller ones are hard to handle and trace, and wider ones overlap in Border E, whose footprints are 3 cm apart; the kit's are 26 mm. In every recipe, k is any integer, so the pattern continues both ways.

Border T, slides only: put A at (6k, h). The primitive period is 6 cm. A variant with A at (4k, h) has primitive period 4 cm. A translation must take an A anchor to another A anchor, so no smaller positive slide works. A half-turn would reverse the asymmetric motif; no reversed motif is present.

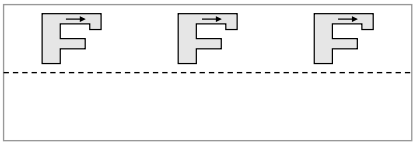
Border G, a glide without its bare reflection: put A at (6k, h) and H A at (6k + 3, -h). A 3 cm rightward glide takes an upper A to the next lower H A and that lower piece to the next upper A. The shortest glide is 3 cm and the shortest slide is 6 cm. Variants G4 (A at (4k, h), H A at (4k + 2, -h)) and G8 (8k and 8k + 4) have shortest glides 2 and 4 cm and shortest slides 4 and 8 cm. A bare horizontal flip lands in the gaps. A 3 cm bare slide lands on the wrong row. No vertical reflection or half-turn matches because it would introduce V A or R A, neither of which occurs.

Border H, a horizontal reflection without any half-turn: put A at (6k, h) and H A at (6k, -h). The middle-line flip matches paired motifs. Every half-turn changes the direction/handedness to R A or V A, forms absent from the border. Horizontal glides by nonzero multiples of 6 cm also work.

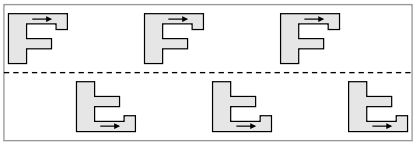
Border R, half-turns without reflections: put A at (6k, h) and R A at (6k + 3, -h). Half-turn about (1.5, 0) matches, and so do half-turns about (1.5 + 3k, 0). The shortest slide is 6 cm. Horizontal or vertical reflection introduces H A or V A, so neither type of reflection matches. No glide works either.

Border E, both perpendicular reflections: repeat four motifs every 6 cm: A at (1.5, h), V A at (-1.5, h), H A at (1.5, -h), and R A at (-1.5, -h). Reflection in y = 0 and in x = 0 both match. Their composition is the half-turn about (0,0). The primitive slide is 6 cm; a 3 cm slide preserves locations but changes the required motif orientations.

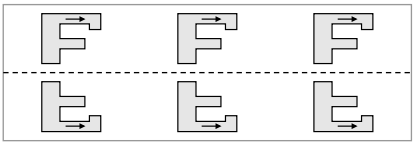
T: slides only



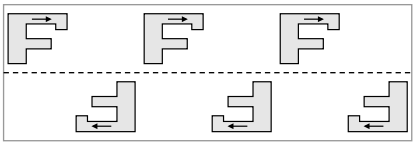
G: a glide, no flip



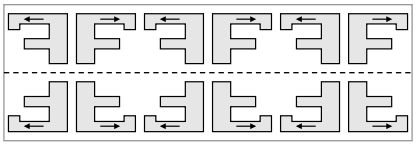
H: the flip, no half-turn



R: half-turns, no flip



E: two crossing flips



The five witnesses, three repeats each, at 30% of actual size: footprints in grey, the middle line dashed, upper footprints centred h above it and lower ones h below. E's vertical mirror lines run midway between neighbouring footprints in each row.

Keys for K-1 and Grades 2-3

K-1

Problem 1: Border T at 4 cm matches after a 4 cm slide and fails after a 2 cm slide. Keep the original fixed and compare several repeats; children may compare distances without measuring.

Problem 2: Border T with 4 cm spacing and with 6 cm spacing gives two different shortest slides. The child need not name centimeters; compare the moved copies. Repeating a 4 cm block twice does not make an 8 cm primitive period.

Problem 3: Border G is a witness: a flip and 3 cm slide works; the flip alone fails. It is enough for a child to demonstrate those two facts on several repeats and describe the continuation.

Problem 4: G8 matches after the flip and a 4 cm slide, where G used 3 cm; the flip alone still fails. Use a long strip to see three repeats. Any glide border with a longer repeat than the child's Problem 3 border will do.

Problem 5: Start with Border H. Shift every lower-row mirrored footprint 3 cm right to obtain Border G; the old middle-line reflection now fails. Repeat the same edit in every 6 cm block so the new border still repeats.

Problem 6: Border R matches after a half-turn about (1.5,0). Border T has no matching half-turn about any point, because no half-turned footprint occurs anywhere. Testing just a printed tick as a center is not sufficient.

Problem 7: Answers depend on the traded borders. Use the recipes on page 4 as checks, not as a required catalog. Encourage the tester to show a matching move and a failed move, with the whole repeat accounted for.

Grades 2-3

Problem 1: Border T at 4 cm and 6 cm spacings. Mark the displacement of a particular motif on the tracing copy. Its image must be the nearest next A for the shortest positive match.

Problem 2: Border G and G4 (page 4), with shortest slides 6 and 4 cm; both fit the boxes. In each, neither the bare horizontal flip nor the bare half-length slide works, although their glide does.

Problem 3: On every border satisfying Problem 2, the shortest glide is half the shortest slide: 3 and 6 cm in G, 2 and 4 cm in G4. Two borders show a pattern, not one coincidence; it is a theorem under the stated assumptions, proved on page 7.

Problem 4: Border H has the requested flip and no half-turn about any point. The asymmetric motif rules out the reversed orientations a half-turn would need.

Problem 5: Border R works. Its middle-line flip fails. The task does not demand the absence of all other reflections, but this witness has none.

Problem 6: One simple edit is H to G above, breaking the old reflection and keeping period 6. To break an old translation instead, start with T at $x = 6k$. In every 12 cm block move the footprint at $x = 12k$ to $x = 12k + 0.5$, leaving the one at $x = 12k + 6$ unchanged. The new gaps alternate 5.5 and 6.5 cm. The old 6 cm slide fails and a 12 cm slide works. This edit uses the explicitly allowed larger repeating part; it is not a one-off defect in an otherwise endless unchanged border.

Key for Grades 4-5

Problem 1: Border T with periods 4 and 6 cm. Since each motif is asymmetric, a matching translation preserves its form and takes an anchor to an anchor. Consecutive anchors are precisely 4 or 6 cm apart, excluding smaller positive translations. An unnecessarily large chosen repeat is not evidence of a large primitive period.

Problem 2: Border G has shortest slide 6 cm and shortest glide 3 cm. Tracking one upper A forces slide displacement into 6 times an integer, or glide displacement into 3 plus 6 times an integer. Each permitted displacement also maps the lower mirrored row correctly.

Problem 3: A 3 cm glide is the flip over the middle line followed by a 3 cm slide along it, as Problem 2 defines it. A 6 cm slide must work: apply the given glide twice. A 3 cm slide need not work; Border G is a counterexample. Do not infer that 6 is necessarily the shortest matching slide, since the hypothesis only gives one matching glide.

Problem 4: Border E is a construction. The perpendicular flips force a half-turn at their intersection. As the border already repeats, composing these flips and the period generates further symmetries: translated copies of the perpendicular mirror line, intermediate perpendicular mirror lines, half-turn centers, and glides formed from the middle-line flip and translations. For E, reflections occur in $y = 0$ and in $x = 3k$, half-turn centers at $(3k,0)$, slides by $6k$, and glides along $y = 0$ by nonzero $6k$. The two crossing flips alone do not produce a nonzero translation; the periodicity is a separate given feature.

Problem 5: Border R matches by a half-turn and has neither a middle-line reflection nor any perpendicular reflection. At an upper A, either type of reflection would require one of the missing mirror forms. The lack of those forms rules out every possible perpendicular axis, not only the axis through the displayed center.

Problem 6 (its own page, with room to write): answers are construction-dependent. "Explain why no other move matches" asks for the completeness argument below in a child's form: follow one footprint, which every matching move must send to a footprint of the same form. A complete answer for G is: translations by $6k$ cm ($k \neq 0$ for a moving slide), glides along $y = 0$ by $(6k + 3)$ cm, no pure reflection, and no nonidentity rotation. Include the identity only if discussing the full group. For R: translations by $6k$ cm and half-turns about $(1.5 + 3k,0)$, with no reflections or glides. Negative displacements are included.

Why these descriptions are complete

For these nonempty bounded-width discrete borders, the unbounded direction is the horizontal direction. An isometry preserving the border must preserve that direction, so its linear part is one of the four sign changes: identity, horizontal reflection, vertical reflection, or a half-turn. A nonzero vertical translation is impossible for a bounded-width pattern. The symmetric two-row examples force the line or centers to have $y = 0$. The asymmetric motif then forces the listed orientation, and its anchor forces exactly the stated x offsets. This is an adult completeness argument; children can explain the corresponding anchor-and-direction obstruction concretely.

Optional extension: Compare H and G. Both have matching glides, but only H has the bare reflection. Ask which shortest-distance relationship changes and why. Seven frieze names are not needed to settle this question.

Proofs and source notes

Shortest glide theorem

The translations form P times the integers. Suppose G_a is a symmetry along y = 0. Because H and horizontal translations commute, $G_a^2 = T_{2a}$, so 2a is an integer multiple of P. Composing G_a with any matching translation changes its displacement by a multiple of P. Reduce a modulo P into [0,P). The only possibilities are 0 and P/2. If the remainder were 0, composing with the inverse translation would make H itself a symmetry. Thus when H fails, the remainder is P/2 and the least absolute nonzero glide displacement is P/2. Any two glides along the same axis differ by a matching translation, so all their displacements are (k + 1/2)P.

If H is a symmetry, composing G_a with H gives T_a; hence glide displacements are precisely translation displacements. Conversely, every translation composed with H gives a glide when the displacement is nonzero. Thus the shortest nonzero glide length is P in that case.

What the finite checks establish

The independent guide check represents the exact notched polygon by integer-coordinate vertices and verifies that no nonidentity cyclic rotation or reversal of its vertices preserves all pairwise distances. It then checks each construction by exact positions and oriented motif type in a periodic cell, including the larger-repeat edit. This verifies these recipes and their primitive periods. The endless symmetry statements follow from their repetition rules and the geometric argument on page 6, not from a finite picture's endpoints.

Important teaching distinctions: a symmetry of a footprint is not the same as a symmetry of the arrangement; a matching glide need not split into two matching component moves; a long drawn repeat need not be primitive; and failure around one chosen center does not rule out every half-turn.

Sources

Joseph A. Gallian, **Contemporary Abstract Algebra**, author-hosted Chapter 28, **Frieze Groups and Crystallographic Groups**, pp. 446-449, especially pattern II, its squared-glide translation subgroup, and pattern VII's perpendicular reflections. Author-hosted chapter. This source supplies the symmetry-group context; the exact notched motif, constructions and shortest-period proof here are independently developed. The full seven-type classification is background only.

Laura Givental, Maria Nemirovskaya and Ilya Zakharevich, **Math Circle by the Bay**, AMS/MSRI Mathematical Circles Library 21 (2018), Preface pp. viii-ix. It describes substantial themes, manipulatives, dialogue and varied pace. The proposed one-hour sequence, exact motif preparation and readiness gates here are local plans, not a report of testing these tasks in that source.

Packet basis: current Week 35 Footprint borders, October 23, 2026: K-1 v4, Problems 1-7; Grades 2-3 v4 and Grades 4-5 v4, Problems 1-6. K-1 skips v3 because the October 4 guide called the packet printed W35-k-1-v2 "K-1 v3", and an archived six-problem packet also printed W35-k-1-v2. This guide uses the revised shortest-period wording, freely chosen half-turn centers, excluded guide lines and enlarged-repeat edit convention.

Starting routes and return visits

Planning update of 4 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

Use and return choices

K-1 first route: Problem 1 makes one repeating border and tests matching and failing slides with a tracing copy; Problem 2 compares two shortest slides when the first action is reliable. Keep at least three or four visible repeats and the original fixed. Introduce the combined flip-and-slide only before Problems 3-4. On return, K-1 can change a matching flip (P5) or investigate half-turns and partner testing (P6-7). Grades 2-3 can choose reflection or half-turn constructions and edits (P4-6). Grades 4-5 can return to forced compositions (P3-4), or spend a visit finding all symmetries of one border (P5-6).